

R E E L

Netflix Recommendation System for Personalized Content Discovery

Learning user preferences from 24 million ratings to surface the right films — and explaining why.

24M

ratings analyzed

0.8146

best RMSE

0.8497

best MAP@10

Discovery is a ranking problem, not just a prediction problem

Every day, users face an overwhelming catalogue. The challenge is not rating a film they've already chosen — it's **surfacing relevant titles they haven't seen yet**.

This reframing matters: it makes **ranking quality** — putting the right items at the top — as important as raw accuracy. That single idea drives every modelling choice in this project.

We use the Netflix Prize dataset and stay strictly within the provided data — no external metadata.

RMSE

Rating accuracy

How close are predicted ratings to the true ones?

MAP@10

Ranking quality

Are the right items ranked at the top of the list?

Three paradigms, one honest evaluation

01

Weighted-Mean

Non-personalized baseline. Damped average rating (IMDb formula). Same list for everyone — the bar to beat.

02

Matrix Factorization

Latent factors via SGD, from scratch. Learns taste affinity per user & movie. The Netflix-Prize workhorse.

03

Item-based k-NN

Item-item cosine similarity. Slightly less accurate, but interpretable — powers the explanations.

EVALUATION

Temporal split

Hold out each user's newest 20% — no future leakage

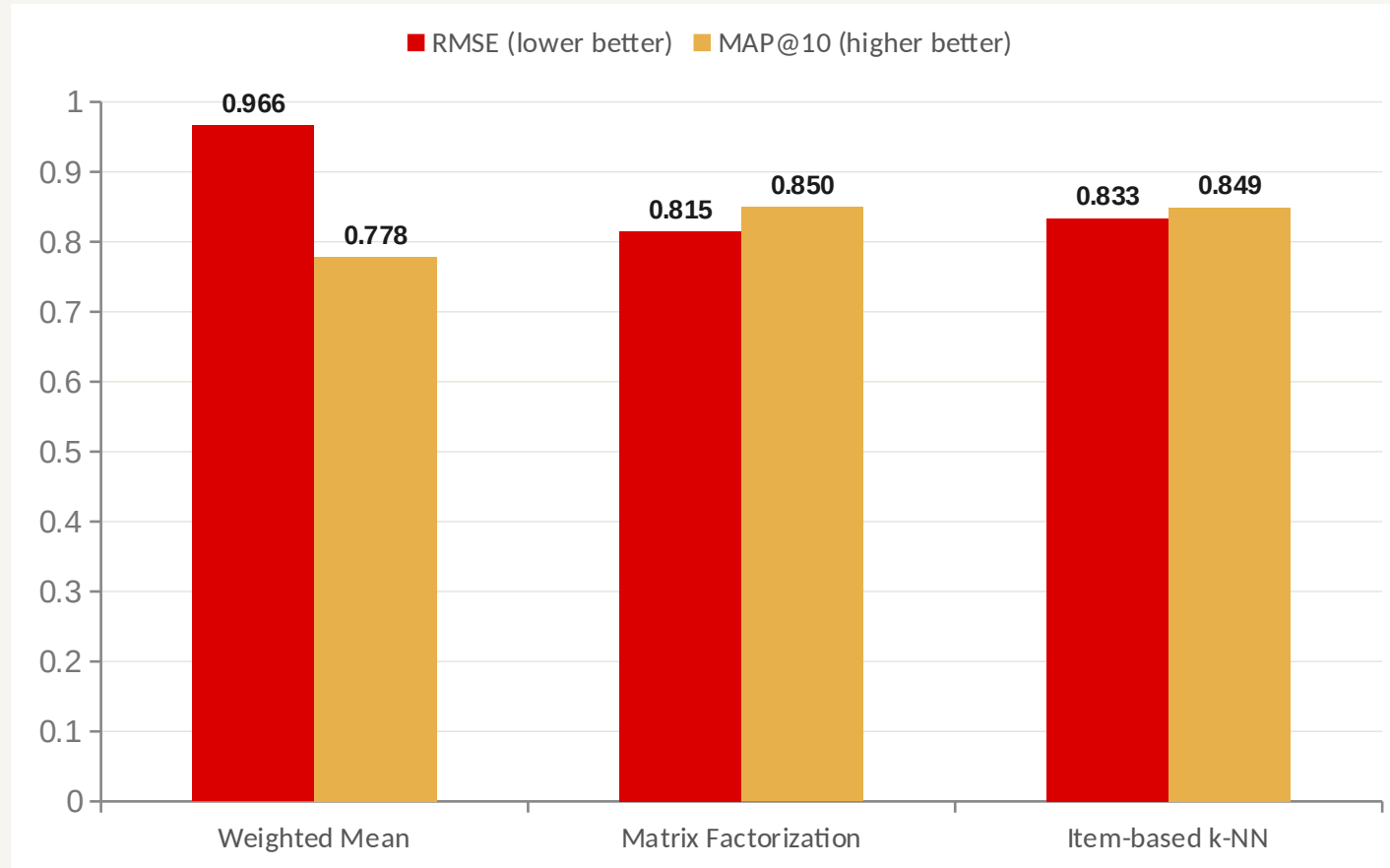
Two metrics

RMSE (accuracy) + MAP@10 (ranking)

Relevance ≥ 3.5

Definition used for MAP@10 ranking

Matrix Factorization wins on both metrics



Final results (held-out test set)

Model	RMSE	MAP@10
Weighted Mean	0.9661	0.7784
Matrix Factorization	0.8146	0.8497
Item-based k-NN	0.8334	0.8488

vs. Netflix Cinematch

0.8146 beats **0.9474**

Our best model is comfortably below Netflix's historical benchmark.

● KEY INSIGHT

RMSE and ranking quality **are not the same goal**

The Weighted-Mean baseline posts a not-unreasonable RMSE (0.9661) — but its MAP@10 collapses to 0.7784, far below the personalized models (~0.849).

Why? A non-personalized model hands every user the **identical list** — so it cannot order any individual's preferences well, no matter how calibrated its point predictions are.

Weighted Mean — same RMSE-ballpark, but...

0.7784

MAP@10 — weak ranking

Matrix Factorization — personalized

0.8497

MAP@10 — strong ranking

Takeaway: Optimize for the metric that matches the goal. Content discovery is a ranking problem — personalization is what lifts MAP@10.

Every recommendation carries its reason

Matrix Factorization ranks the items; Item-k-NN supplies the “because you watched” explanation — real output from the run.

User 0 — classics & thrillers

Liked: Jaws, X2, Roger Rabbit

Seven Samurai

↳ because you liked The Godfather, Spinal Tap

North by Northwest

↳ because you liked The Odd Couple, The Godfather

Evil Dead 2

↳ because you liked Nightmare on Elm St, 28 Days Later

User 1 — blockbusters & drama

Liked: LOTR: Fellowship, The Wedding Planner

Batman Begins

↳ because you liked Aladdin, Matrix Revolutions

Ray

↳ because you liked Shrek 2, Something's Gotta Give

Secondhand Lions

↳ because you liked Pay It Forward, 50 First Dates

User 2 — feel-good & romance

Liked: Bend It Like Beckham, What Women Want

Finding Neverland

↳ because you liked Bend It Like Beckham, About a Boy

The Godfather

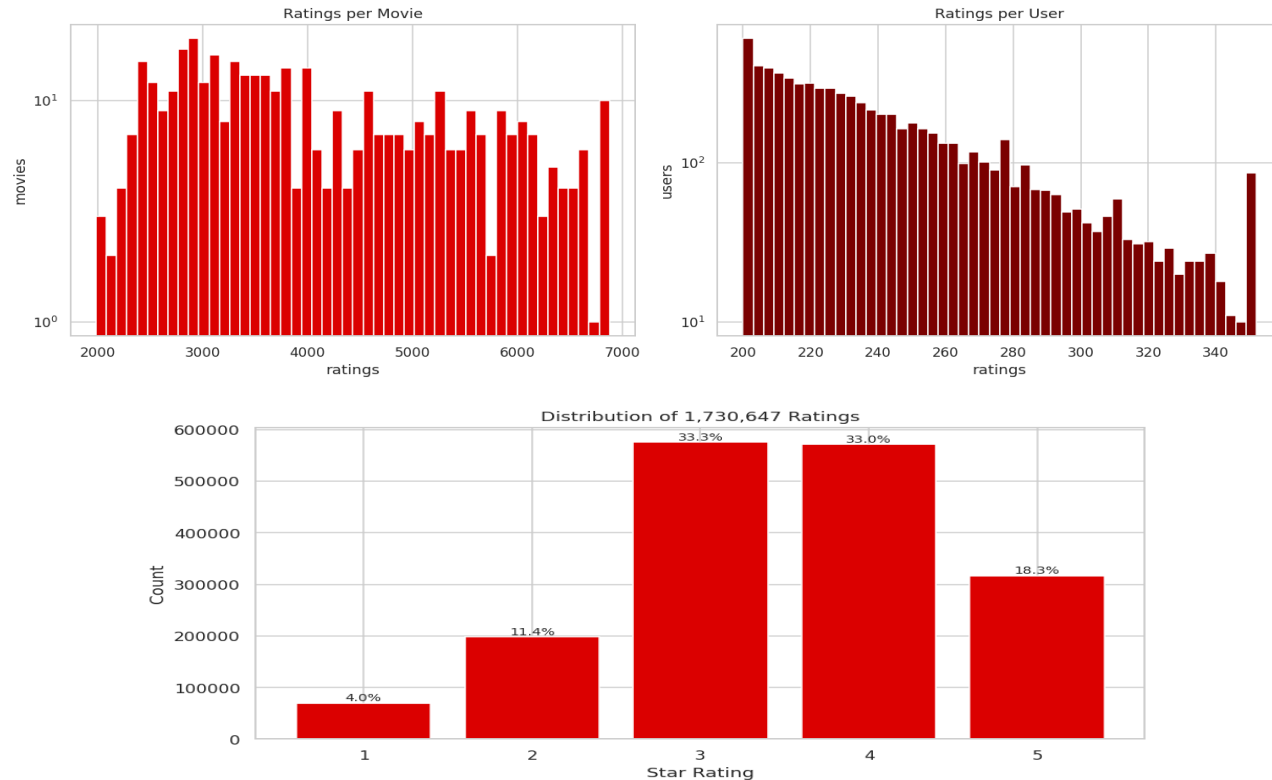
↳ because you liked Silence of the Lambs, Jaws

The Lion King

↳ because you liked Finding Nemo, Ghost

● THE DATA

24 million ratings, shaped by steep long tails



24,053,764

ratings parsed from one data file

1,730,647

ratings after smart subsetting

7,198 × 419

users × movies in working set

~16%

of ratings held by top 10% of movies

Insight: popularity-only models recommend the same hits to everyone — weak discovery.

● SUMMARY

What we built — and what it shows

● **Three models, three paradigms**

Weighted-Mean, Matrix Factorization, and Item-k-NN — compared head-to-head.

● **Best result beats the benchmark**

Matrix Factorization: RMSE 0.8146, MAP@10 0.8497 — under Cinematch's 0.9474.

● **A leakage-free evaluation**

Temporal split mirrors the real task; RMSE + MAP@10 measure different things.

● **The core finding**

Accuracy $\neq$ ranking. Personalization is what makes content discovery work.

● **Explainable by design**

MF ranks, k-NN explains — every recommendation carries a reason.

Future work: scale to all 100M ratings · SVD++ & temporal dynamics · coverage/diversity metrics · cold-start handling.